Binary Distillation

Lab Report
Unit Operations Lab #2
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Group 2 of the Tuesday AM section

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Executive Summary

In this lab, a binary batch distillation column was utilized to distill a mixture of ethanol and water at 12.04wt% for experiment A and 10wt% mixture for experiment B. Distillation is a significant separation process in chemical engineering that uses vapor pressure of the components mixed to separate them into liquid and gas. Due to water having a lower volatility than ethanol, it was present in higher concentrations in the liquid phase. Which then meant that the ethanol was present in higher concentrations in the gas phase. In distillation, gravity separates the phases, liquid falls, and gas rises resulting in a countercurrent flow through the column. Such a flow creates a better separation than a concurrent process which is why the countercurrent flow is more ideal in this lab.

Experiment's A objective was to analyze the effects of boil up rate variation on pressure drop over the column and the column's efficiency. This was done by measuring the manometer's pressure drop at different boil up rates. Also, the bottom and top liquid samples were collected across different boil up rates to determine the ethanol composition. As the boil-up ratio increases the pressure drop had a logarithmic behavior. It increased then eventually started to level out.

Experiment's B objective was to carry out distillation at two different constant reflux ratios. This varied the bottom and top compositions over time. The bottom and top liquid samples were collected at different times for each individual reflux ratio, 5:1 and 3:1. The difference between the bottom composition was not significant and the distillate composition varied constantly by about 4% with the 3:1 ratio being higher than the 5:1 ratio.

The data satisfied the expected trends and correlated to the relevant theory. The pressure drops had a logarithmic behavior as the boil up rate increased with the column efficiency of 36.8%. It was then observed that the higher reflux ratio resulted in a better separation of the components. This is seen in the increase of ethanol weight percent in the distillate fraction.

Introduction

The objectives are to determine the variation with boil-up rate of pressure drop over the distillation column, determine overall column efficiency and to vary the top and bottom compositions at various reflux ratios. The boil-up rates were determined by operating at total reflux and measuring the different boil-up rates. The overhead and bottom liquid samples were taken at different boil-up rates to determine the ethanol composition via refractive index and the overall column efficiency was calculated [1].

The apparatus consists of a distillation column with eight sieve plates, condenser, reboiler, heater, valves, eight thermocouples and a control panel. Distillation columns are widely used in industry to separate mixtures, refine oil, desalination of water, create liquor, beer and wine, and other chemical products [2].

Theory

Distillation is the most used separation method which accounts for more than 90% of the separations in industries and batch distillation is used specifically for alcohol, pharmaceuticals and oil distillation. [3] Distillation depends on the boiling points of the components that are to be separated so that the more volatile component can be boiled up and purified. (Trayed) columns have trays so that there is more contact between the vapor and liquid phases which results in a better separation of the components. These (sieve) trays have holes in them so that the vapor can rise through the plate and the liquid on it. [4]

Columns also have a reboiler at the bottom to heat up part of the liquid stream back into a vapor phase while the other part becomes the liquid product stream which contains the less volatile component of a binary mixture. While it is not necessary, most distillation columns also have a condenser at the top of the column so that the vapor phase can be cooled down and part of the stream leaves as the overhead product stream which contains the volatile component while the rest of the stream is the reflux stream that is returned to the top plate of the column for further separation. [5] A distillation column is shown below in **Figure 2.1**.

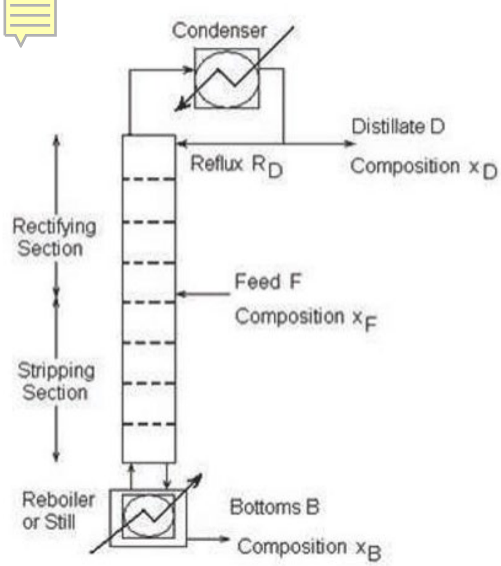

Figure 2.1: Diagram of a Distillation Column

However, in our experiment, we operated a batch distillation column, shown in **Figure 2.2**, which means that the mass/amount of the solution in the system was constant for the most part; it decreased slightly when we would sample the top and bottom stream in order to test the product for the refractive index. The separation of the liquid and gas phases are due to gravity. The vapor phase rises to the top of the column while the liquid is returned to the bottom; thus, the column operates in a counter-current manner. The known values for the vapor-liquid equilibrium of ethanol and water are shown in **Figure 2.3** and this was the graph which we used to plot our experimental data.

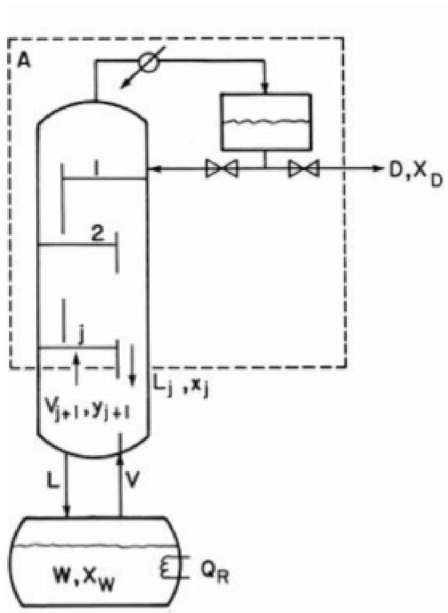

Figure 2.2: Diagram of a Batch Distillation Column

<u>Experiment A:</u>

Increasing the power input to the reboiler increases (controls) the boil-up rate which controls the velocity of the vapors passing through the column. The velocity of the vapors is proportional to the overall pressure drop. A total reflux system means that the instantaneous composition of the distillate keeps changing because the bottom composition of the more volatile component is continuously depleted. The batch system operation was performed at constant reflux to obtain the column efficiency.

<u>Experiment B:</u>

Experiment B was operated under total reflux, which means all recovered ethanol was returned to the column for further distillation cycles. Experiment B deals with changing the reflux ratio or amount of ethanol cycled back to the column. Reflux ratio is defined as the amount of distillate returned to column versus amount of distillate recovered. This is achieved by changing the time interval at which solution is returned and recovered. Increasing the reflux ratio increases the purity of the distillate recovered the tradeoff however is there would be an increase in the amount of time to recover similar amounts of distillate at lower reflux ratios. Reflux ratios under consideration for Experiment B are 5:1 and 3:1.

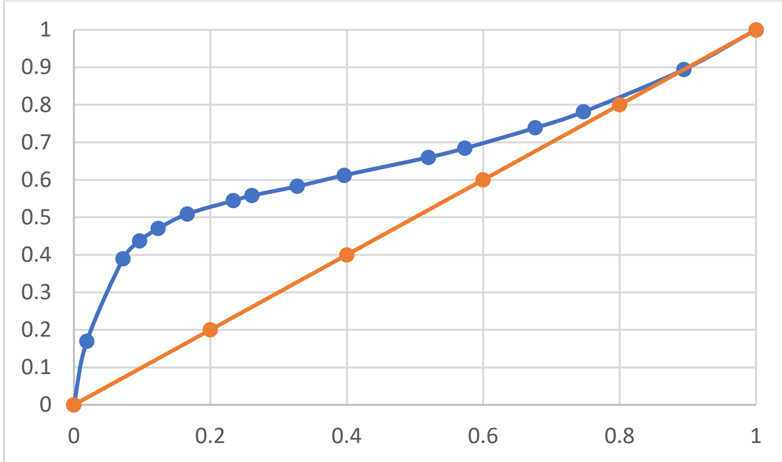

Figure 2.3: Vapor-Liquid Equilibrium for Ethanol and Water at 1atm

Results

Experiment A

The objective of this experiment was to analyze the operational behavior of a batch distillation under total reflux conditions. The column operates under two modes: unsteady and steady state. Prior to beginning Experiment A, the refractometer was calibrated in order to correlate a relationship between the refractive index and the ethanol composition of the sample. The polynomial best fit line was used for the calibration curve shown in **Figure 2.4** below.

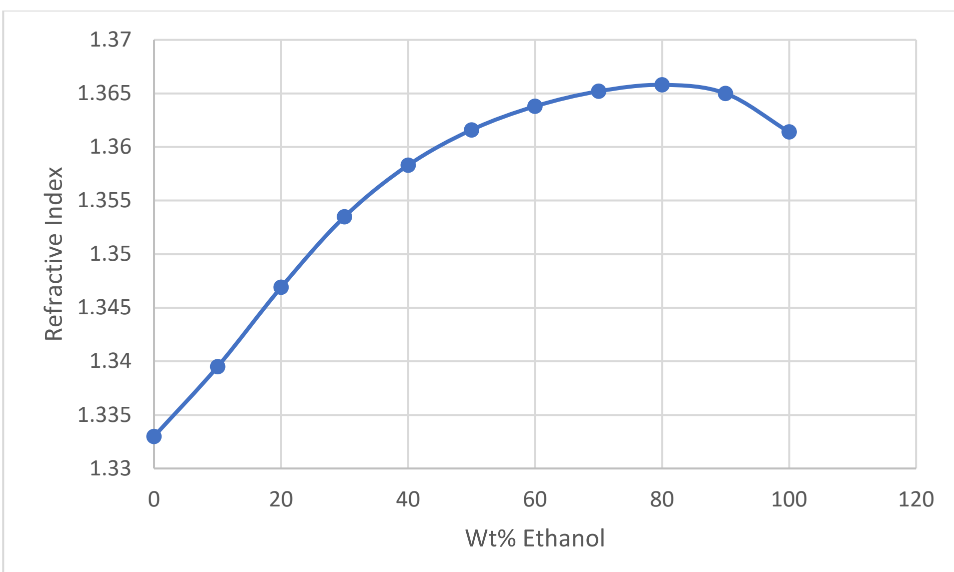

Figure 2.4: Refractive Index vs wt% Ethanol

Initially, we set the power to 1.50 kW to get our solution to the boiling point of ethanol which is approximately 78 °C, and then we reduced the power as the column was approaching steady state. From **Figure 2.5** it was noted that the column progresses linearly in unsteady state until it reaches steady state conditions soon after the 20-minute period from power startup.

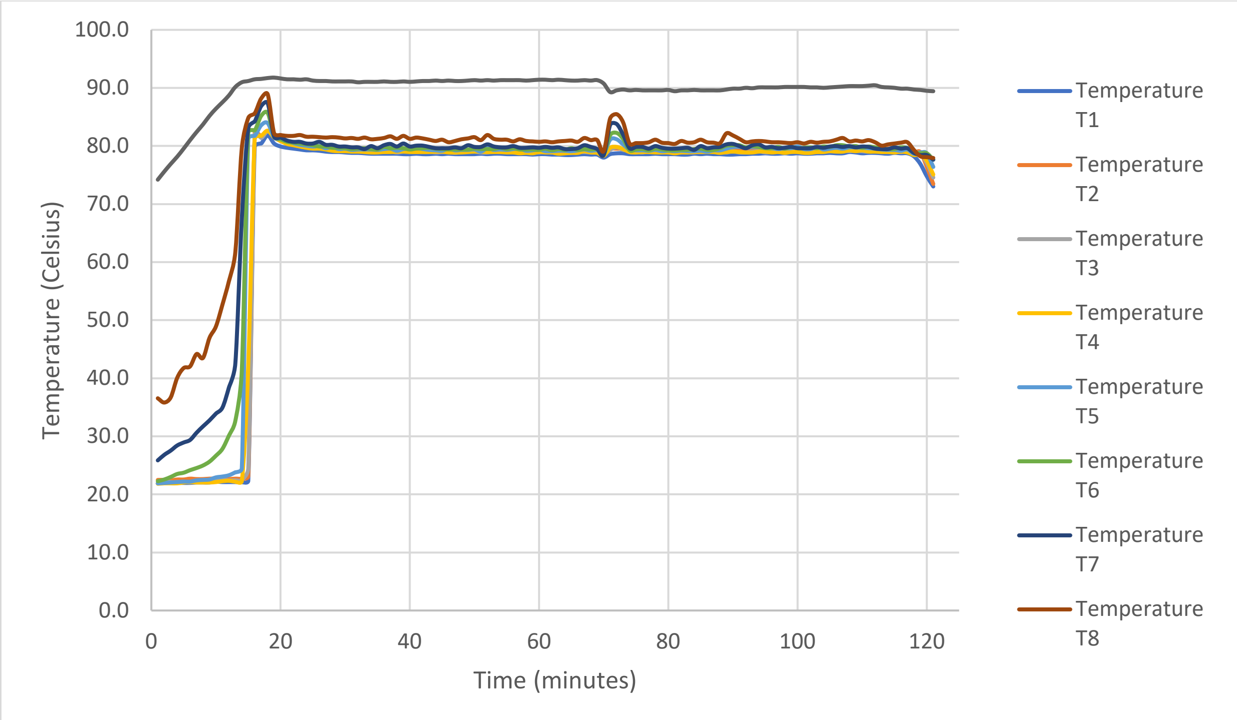

Figure 2.5: Distillation Stage Temperatures vs. Time (unsteady to steady state)

The pressure drop across the column was measured with the manometer. Then the boil-up rate was calculated from the measured volume retrieved at the top of the column. We were able to do this because the column was operating at total reflux. We measured the time it took to accumulate the collected volume, and then we calculated the boil-up rate. **Figure 2.6** shows the pressure drop versus boil-up rate. The third point on **Figure 2.6** indicates that the boil-up rate is affected significantly by a greater pressure drop in the column.

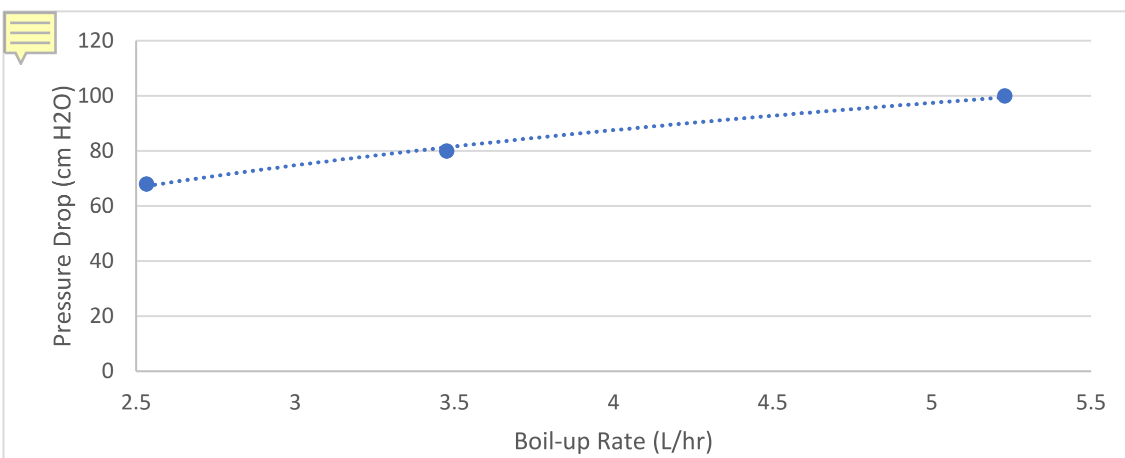

Figure 2.6: Boil-up Rate vs. Pressure Drop Graph

Experiment B

The objective of this experiment was to run the distillation column at two constant reflux ratios. The system was applied with 0.75kW of power and ran at total reflux until equilibrium was reached. Upon reaching equilibrium, the reflux was changed from total to the first ratio of inspection. Samples of both the top and bottom products were retrieved and analyzed, after cooling. Changing the reflux ratio caused the top and bottom compositions to change as time progressed. This cycle was repeated for the second reflux ratio. The average temperature of the system was approximately 80 °C, which is higher than the boiling point of ethanol, and the data from this experiment is shown in **Figure 2.7** to **Figure 2.9**.

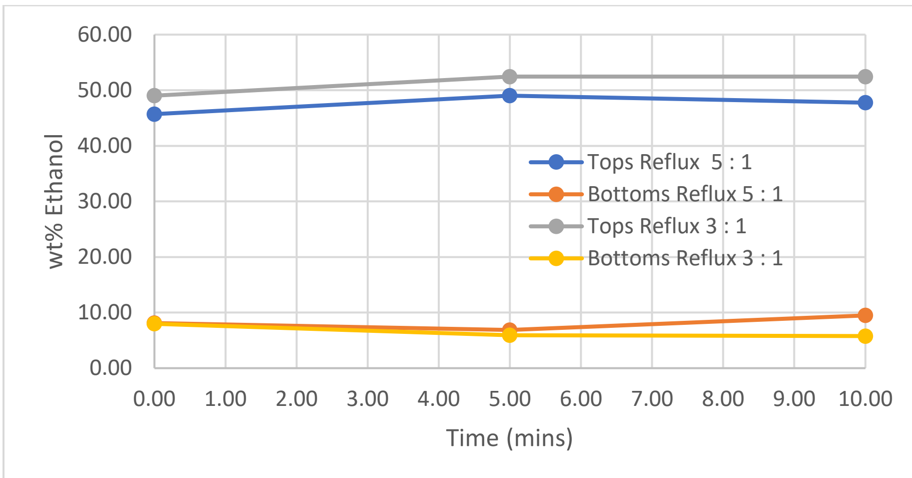

Figure 2.7: Change in wt% over time for constant reflux ratios

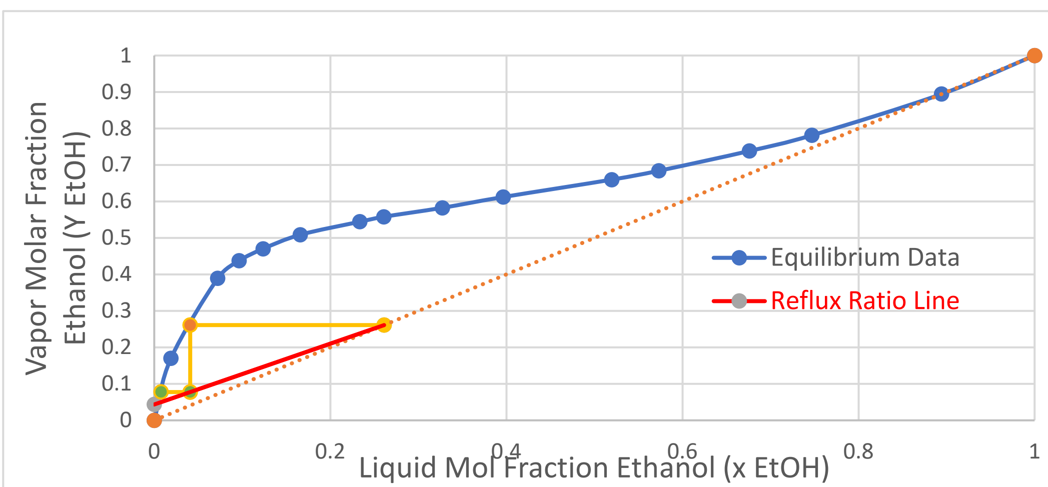

Figure 2.8: Trays plotted on McCabe-Thiele Diagram for 5:1 Reflux Ratio

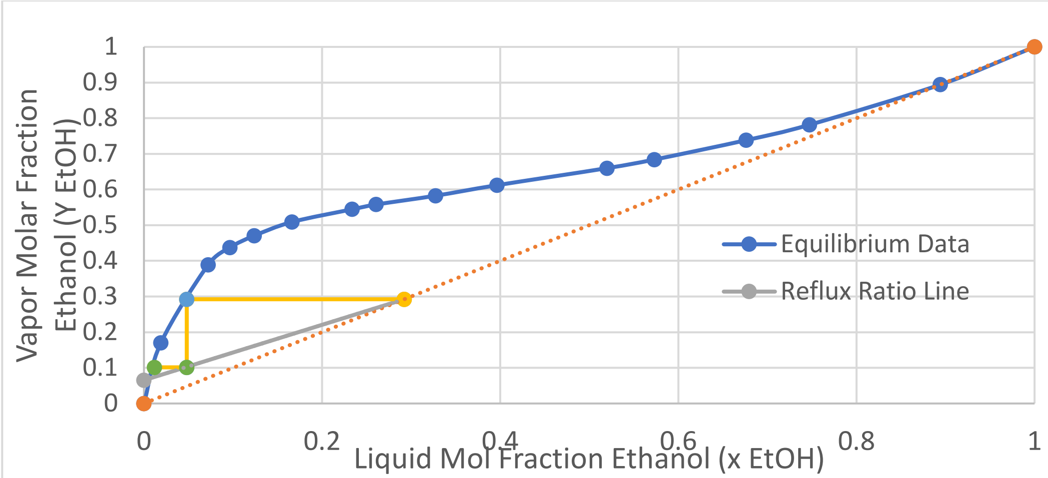

Figure 2.9: Trays plotted on McCabe-Thiele Diagram for 3:1 Reflux Ratio

Discussion

In this experiment, a binary distillation column was analyzed for the separation of ethanol and water. A binary mixture of 12.04 wt.% ethanol in water was separated for Experiment A. Some parameters, such as the boil-up rate increased, but time decreased as the power provided to the distillation column was increased. Due to the power supplied to the distillation column increasing, the heat delivered to the water and ethanol solution increased and thus it reached the boiling point faster. This causes the ethanol molecules in the solution to transfer/jump into the vapor phase more quickly and a larger volume is boiled in the same amount of time (boil-up ratio) when comparing Trial 3 to Trial 1. Because of this reason, it also took less time to collect the same volume of sample for Trial 3 than for Trial 1. Another notable trend in this experiment is that refractive index of the distillate gets closer to 1.3614, which is the refractive index of pure ethanol, and that the bottoms is closer to 1.3330, which is the refractive index of water. This means that the desired separation has taken place in column.

For part B, the distillation was carried out with two reflux ratios, 5:1 and 3:1. Both reflux ratios follow the same trend in that the top product increases in ethanol concentration but then decreases, and the bottom product initially decreases and then increases in ethanol concentration. One thing to notice is that the 3:1 reflux has a larger change in ethanol concentration than the 5:1 reflux ratio. When the reflux ratios were 5:1 and 3:1, the number of theoretical trays were 1.23 and 1.44 respectively.

Conclusion

The separation of a binary mixture in a distillation column agreed with the theory. The boil-up rate increases with higher power supply and is more noticeable with large pressure drop. As the ethanol percentage increased in relation to water, it had a higher refractive index as shown in **Figure 2.4**. These values were used to compare the data from experiment A. It helped calculate the compositions of the distillate and bottoms. Once our system reached steady state, samples were collected from tops and bottoms. From the compared data, we were able to determine the weight and mole percent. For experiment B, samples from the distillate and bottoms were tested at two reflux ratios, 5:1 and 3:1. In **Figure 2.8** and **2.9** we compared our data results with the standard McCabe Thiele Diagram and it showed that our theoretical number of stages, 3, deviates from the actual 8 stages. These results yielded a low efficiency of 37%.

References

[1] ChE 382: Unit Operations Laboratory. Binary Distillation. 2019.

[2] Sylvan, R. "What Are the Uses of Distillation in Industry?" *BizFluent.* Sep 2017. https://bizfluent.com/about-5941994-uses-distillation-industry-.html

[3] Wankat, P., *Separation Process Engineering*. Upper Saddle River: Prentice Hall, 2012.

[4] "Distillation Column Internals." *Distillation Column: Column Internals, Bubble Cap Trays, Valve Trays, Sieve Trays, Structured Packing*, Werner Sölken, www.wermac.org/equipment/distillation_part2.html.

[5] "What Are Trays and How Are They Used in Process Plants?" *Amacs Process Towers Internals*, 20 Sept. 2018, www.amacs.com/what-are-trays-and-how-are-they-used-in-process-plants/.

Appendix

Appendix I: Data Tabulation/Graphs:

Table 7.1 Calculation Data Trial 1 Part A

Trial 1	
Power, KW	0.75
Pressure Drop, cm H2O	68
Volume, ml	11.16
Time, sec	11.18333
Boil-up Rate, L/hr	3.592489
Refractive Index	
Distillate	Bottom
1.3618	1.3367
1.3615	1.3378
1.3621	1.3347

Table 7.2 Calculation Data Trial 2 Part A

Trial 2	
Power, KW	1
Pressure Drop, cm H2O	80
Volume, ml	11.21
Time, sec	8.26
Boil-up Rate, L/hr	4.885714
Refractive Index	
Distillate	Bottom
1.3621	1.3431
1.3658	1.3431
1.3615	1.3484

Table 7.3 Calculation Data Trial 3 Part A

Trial 3	
Power, KW	1.25
Pressure Drop, cm H2O	100
Volume, ml	11.38
Time, sec	5.533
Boil-up Rate, L/hr	7.404301
Refractive Index	
Distillate	Bottom
1.364	1.3418
1.3623	1.3443
1.3649	1.3423

Table 7.4 VLE data for EtOH and H2O at 1atm y and x in mole fractions

X_{etoh}	X_w	Y_{etoh}	Y_w	T, °C
0	1	0	1	100
0.019	0.981	0.17	0.83	95.5
0.0721	0.9279	0.3891	0.6109	89
0.0966	0.9034	0.4375	0.5625	86.7
0.1238	0.8762	0.4704	0.5296	85.3
0.1661	0.8339	0.5089	0.4911	84.1
0.2337	0.7663	0.5445	0.4555	82.7
0.2608	0.7392	0.558	0.442	82.3
0.3273	0.6727	0.5826	0.4174	81.5
0.3965	0.6035	0.6122	0.3878	80.7
0.5198	0.4802	0.6599	0.3401	79.7
0.5732	0.4268	0.6841	0.3159	79.3
0.6763	0.3237	0.7385	0.2615	78.74
0.7472	0.2528	0.7815	0.2185	78.41
0.8943	0.1057	0.8943	0.1057	78.15
1	0	1	0	78.3

Table 7.5 Ethanol Distillate Table 7.6 of Bottom

Compositions of Ethanol Compositions

Distillate Ethanol Composition	
Trial 1	0.5283
Trial 2	0.5859
Trial 3	0.6199
Trial Average	0.578033
Xd EtOH	0.348815
Xd H2O	0.651185
Yd EtOH	0.581935
Yd H2O	0.418065
Relative Volatility (α) Distillate	0.384822

Bottom Ethanol Composition	
Trial 1	0.0626
Trial 2	0.1743
Trial 3	0.1452
Trial Average	0.127367
Xb EtOH	0.053993
Xb H2O	0.946007
Yb EtOH	0.31573
Yb H2O	0.68427
Relative Volatility (α) Bottom	8.08441

Table 7.7 Efficiency Calculations

Efficiency Calculations	
Average Relative Volatility	1.763819817
Number of Theoretical Tray	2.945766328
Actual Number of Trays	8
Efficiency	36.82207909

Appendix II: Error Analysis:

Table 7.8 Standard Deviation samples for Bottom and Distillate of each trial

		Distillate		Bottoms	
		Refractive Index	Standard Deviation	Refractive Index	Standard Deviation
	Sample 1	1.3618		1.3367	
	Sample 2	1.3615		1.3378	
	Sample 3	1.3621		1.3347	
Trial 1	Average	1.3618	0.000244949	1.3364	0.001283225
	Sample 1	1.3621		1.3431	
	Sample 2	1.3658		1.3431	
	Sample 3	1.3615		1.3484	
Trial 2	Average	1.3631333	0.001901461	1.3448667	0.002498444
	Sample 1	1.364		1.3418	
	Sample 2	1.3623		1.3443	
	Sample 3	1.3649		1.3423	
Trial 3	Average	1.3637333	0.001078064	1.3428	0.001080123

There were a number of experimental errors throughout the entire process that if corrected would've led to more accurate values. The first of which being the preparation of the ethanol solution. In the pre-lab, we recorded the volume of Ethanol to be used as the mass of the ethanol to be added, which lead to an Ethanol solution that was 12.04 wt.% instead of 10 wt.% according to the lab manual. The second and probably most important error was the solution we lost to incorrectly 'closing' the bottom after retrieving our sample. The valve was left open and a majority of the solution was lost to the drain which caused the distillation columns safety protocol to be activated and the entire system was powered down. In an attempt to salvage the remaining solution and continue with that portion of the experiment, additional ethanol solution was prepared. We took the refractive index of the bottom solution and measured the wt.% of the Ethanol at the bottoms. When compared to the refractive index calibration curve, it was determined that the bottom solution was 19 wt. %. It was determined later the calibration curve would need to be developed again to obtain more accurate results. An additional 4L of 19 wt. % ethanol was added to the batch. Subsequent trials were run with a mixture of the remaining bottoms solution and the second ethanol solution produced. There were deviations in the concentration between the two solutions used. The solution for experiment A had to be reheated after the system shut down and took much longer, so Experiment B was completed a day later with the correct weight percent (10 wt. %). For Trial 1 of experiment B, the system wasn't given enough time to reach steady state before a sample was taken. The initial sample was discarded and not used in our calculations, but the sample may have disturbed the mass flow rate and equilibrium.

Appendix III: Sample Calculations:

Find $\boldsymbol{\alpha_d}$:

$$Relative\ Volatility = \alpha_{EtOH}$$
$$Vapor\ Mole\ Fraction\ of\ Ethanol = y_{EtOH}$$
$$Liqui\ Mole\ Fraction\ of\ Ethanol = x_{EtOH}$$
$$Vapor\ Mole\ Fraction\ of\ Water = y_{H_2O}$$
$$Liquid\ Mole\ Fraction\ of\ Water = x_{H_2O}$$

$$\alpha_d = \frac{\frac{x_{EtOH}}{y_{EtOH}}}{\frac{x_{H_2O}}{y_{H_2O}}}$$

$$\frac{\left(\frac{0.34881}{0.65118}\right)}{\left(\frac{0.41806}{0.58193}\right)} = 0.384822$$

Find $\boldsymbol{\alpha_b}$:

$$Relative\ Volatility\ in\ the\ Bottom = \alpha_b$$

$$\alpha_b = \frac{\frac{y_{EtOH}}{x_{EtOH}}}{\frac{y_{H_2O}}{x_{H_2O}}}$$

$$\frac{\frac{0.31573}{0.05399}}{\frac{0.68427}{0.94600}} = 8.08441$$

Find $\boldsymbol{\alpha_{avg}}$:

$$\alpha_{avg} = \sqrt{\alpha_d * \alpha_b}$$

$$\alpha_{avg} = \sqrt{.384822 * 8.08441} = 1.76382$$

Find theoretical plates, n:

$$n + 1 = \frac{\log\left[\left(\frac{x_{EtOH}}{x_{H2O}}\right)_d \left(\frac{x_{H2O}}{x_{EtOH}}\right)_b\right]}{\log\left(\alpha_{\frac{EtOH}{H2O}}\right)_{avg}}$$

$$n + 1 = \frac{\log\left[\left(\frac{0.34881}{0.65118}\right)_d \left(\frac{0.946007}{0.053993}\right)_b\right]}{\log\ (1.76382)}$$

$$n = 3.945766 - 1$$

$$n = 2.945766$$

Find column efficiency:

$$E = \frac{number\ of\ theoretical\ plates}{number\ of\ actual\ plates} * 100\%$$

$$E = \frac{2.945766}{8} * 100\%$$

$$E = 36.82\%$$

Equation for Reflux Ratio:

$$y = \frac{R}{R+1}\, x_n + \frac{1}{\mathrm{R}+1} x_D$$

Equation for 5:1 Reflux Ratio:

$$\mathrm{y} = \frac{5}{6} x_n + \frac{1}{6}(0.2611)$$

Equation for 3:1 Reflux Ratio:

$$\mathrm{y} = \frac{3}{4} x_n + \frac{1}{4}(0.2923)$$

Appendix IV: Individual Team Contributions:

NAME: Evelyn Almonte

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Used two days for lab
Pre-Lab Editing	1	
Experimental Objective (Prelab)		
Apparatus (prelab)	1	
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing		
Executive Summary (final lab)	1.5	
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	4	
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	3	
Error Analysis (final lab)		
Sample Calculations (final Lab)	2	
Power Point Presentation	0.5	
Total	20	

NAME: Sandra Hernandez

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Used two days for lab
Pre-Lab Editing		
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)	1	
Procedure (prelab)	1	
Project Safety Evaluation (prelab)	1	
Final Lab Editing	1	
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)	1	
Results (final lab)	2.5	
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	3	
Error Analysis (final lab)		
Sample Calculations (final Lab)	2	
Power Point Presentation	0.5	
Total	20	

NAME: Tianna Mitchell

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Used two days for lab
Pre-Lab Editing	0.5	
Experimental Objective (Prelab)	0.5	
Apparatus (prelab)		
Materials & Supplies (prelab)	1	
Procedure (prelab)		
Project Safety Evaluation (prelab)	1	
Final Lab Editing	2.5	
Executive Summary (final lab)		
Introduction (Final lab)	1	
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)	2.5	
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)	1.5	
Power Point Presentation	1.5	
Total	20	

NAME: Anne Serban

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Used two days for lab
Pre-Lab Editing	1.5	
Experimental Objective (Prelab)	1.5	
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	1.5	
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)	3	
Results (final lab)		
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	3	
Error Analysis (final lab)	1	
Sample Calculations (final Lab)	1	
Power Point Presentation	0.5	
Total	20	

NAME: Maurice Smith

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Used two days for lab
Pre-Lab Editing		
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)	1	
Project Safety Evaluation (prelab)	1	
Final Lab Editing		
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)	1	
Results (final lab)	3	
Discussion (final lab)	0.5	
References (final and/or prelab)		
Data Tabulation/Graphs (final)	4	
Error Analysis (final lab)		
Sample Calculations (final Lab)	2	
Power Point Presentation	0.5	
Total	20	